Infectious Diseases III: Fungal Infections and Immunizations

All key points for the exam (IDSA mycoses guidelines, CDC/ACIP schedules) — drug tables included

PART A — FUNGAL INFECTIONS

1. Antifungal Pharmacology

Class	Mechanism	Spectrum / key uses
Polyenes (amphotericin B, nystatin)	Bind **ergosterol** in fungal cell membrane → **pores** → leakage (fungicidal)	**Broadest**: Candida, Cryptococcus, Aspergillus, **Mucorales**, endemic fungi • resistant: **C. lusitaniae**, **A. terreus**
Azoles (fluconazole, itraconazole, voriconazole, posaconazole, isavuconazole, ketoconazole, topical azoles)	Inhibit **lanosterol 14-α-demethylase** (fungal CYP450) → ↓ ergosterol synthesis (fungistatic vs Candida)	Varies by agent (see below) • **all inhibit human CYP3A4** → DDIs
Echinocandins (caspofungin, micafungin, anidulafungin, rezafungin)	Inhibit **β-(1,3)-D-glucan synthase** → disrupt **cell wall**	**Candida** (fungicidal, incl. C. glabrata, C. krusei, C. auris) • Aspergillus (fungistatic, salvage) • **no activity** vs Cryptococcus, Mucorales
Flucytosine (5-FC)	Converted to 5-FU → inhibits DNA/RNA synthesis	**Cryptococcal meningitis** (with AmB) • **never monotherapy** (rapid resistance)
Allylamines (terbinafine, naftifine, butenafine)	Inhibit **squalene epoxidase**	Dermatophytes (**onychomycosis**, tinea)
Griseofulvin	Disrupts fungal **microtubules** (mitosis); deposits in keratin	Dermatophytes — **tinea capitis**
Triterpenoids (ibrexafungerp)	Glucan synthase inhibitor (oral)	Vulvovaginal candidiasis

Amphotericin B

- **Formulations: conventional AmB deoxycholate** 0.5–1 mg/kg/day (**max 1.5 mg/kg/day** — fatal cardiac arrest with overdose) • **liposomal AmB (AmBisome) 3–5 mg/kg/day** (mucor 5–10) • **AmB lipid complex (Abelcet)** 5 mg/kg/day • **do not confuse doses** between formulations.
- **Infusion-related reactions** ("shake and bake"): **fever, chills, rigors**, hypotension, nausea → **premedicate**: acetaminophen, diphenhydramine, ± hydrocortisone; **meperidine** for rigors • slow infusion (2–6 h) • less with lipid formulations.
- **Nephrotoxicity** (dose-related; ↑ with other nephrotoxins — aminoglycosides, cyclosporine, vancomycin, NSAIDs) → **sodium loading** (500 mL–1 L NS before and after), use lipid formulations • **hypokalemia**, **hypomagnesemia** (replace), **distal RTA (type 1)**, normocytic **anemia** (↓ EPO), thrombophlebitis.
- Compatible with **D5W only** (precipitates in saline) • monitor SCr, K, Mg daily.
- **Lipid formulations**: less nephrotoxicity and infusion reactions, more expensive • **L-AmB preferred for CNS** and most invasive infections.

Azole antifungals

Agent	Spectrum and uses	Key PK/ADRs/DDIs
Fluconazole	**Candida** (not **C. krusei** — intrinsically resistant; **C. glabrata** dose-dependent/resistant; C. auris often resistant), **Cryptococcus**, Coccidioides • **no mold activity** (Aspergillus, Mucor)	**~100% oral bioavailability** (IV = PO) • **renal elimination** (↓ dose 50% if CrCl ≤ 50) • excellent **CSF and urine** penetration • **moderate CYP3A4**, **strong 2C9/2C19 inhibitor** (↑ **warfarin**, phenytoin, sulfonylureas) • QT, hepatotoxicity • **teratogenic** at high doses (avoid in 1st trimester)
Itraconazole	Candida, **endemic fungi** (histoplasmosis, blastomycosis — drug of choice), Aspergillus (ABPA, chronic), dermatophytes (onychomycosis), sporotrichosis	**Capsules: take with food and acidic drink** (cola) — **avoid PPIs/H2RAs/antacids** • **solution: empty stomach** (better absorbed) • **boxed: heart failure** (negative inotrope — CI in HF) • **strong CYP3A4 inhibitor** • hepatotoxicity • TDM (trough > 1)
Voriconazole	**Drug of choice for invasive aspergillosis** • Candida (incl. C. krusei), Fusarium, Scedosporium • **no Mucorales activity** (breakthrough mucormycosis)	**Visual disturbances** (~30%, transient — blurred, altered color), **hallucinations**, **photosensitivity/skin cancer** (SCC, long-term), **hepatotoxicity**, **QT**, **periostitis** (fluoride, long-term) • **CYP2C19 metabolism** (polymorphisms) → **TDM trough 1–5.5 mcg/mL** • IV vehicle (**SBECD**) accumulates if **CrCl < 50** → use oral • oral on empty stomach • **strong 3A4 inhibitor**: **CI with sirolimus**, rifampin, carbamazepine, long-acting barbiturates, St John's wort, ergots, pimozide • ↓ **tacrolimus by 1/3**, cyclosporine by 1/2
Posaconazole	Broad: Candida, Aspergillus, **Mucorales** (step-down/salvage) • **prophylaxis** in AML/MDS with neutropenia and GVHD	**Delayed-release tablets with food** (300 mg BID day 1 → 300 mg daily) • **suspension needs high-fat meal** (poor absorption; not interchangeable) • **strong 3A4 inhibitor** • QT, hepatotoxicity, **hypokalemia/hypertension** (pseudohyperaldosteronism) • TDM
Isavuconazole (isavuconazonium sulfate)	Aspergillus (= voriconazole, fewer ADRs), **Mucorales**	Prodrug, IV/PO (no SBECD), no food effect • **shortens QT** (CI in familial short QT) • **moderate** 3A4 inhibitor • less hepatotoxicity/visual effects • loading 372 mg (200 mg base) q8h × 6 doses → daily
Ketoconazole (oral)	Rarely used systemically	**Boxed: hepatotoxicity**, QT • **inhibits steroidogenesis** → adrenal insufficiency, gynecomastia (used in Cushing's) • strongest CYP3A4 inhibitor • needs acid • topical forms OK (seborrheic dermatitis, tinea versicolor)

Echinocandins

- **IV only** • **caspofungin 70 mg load → 50 mg daily** (↓ to 35 mg in moderate hepatic impairment; ↑ to 70 mg with inducers like rifampin) • **micafungin 100 mg daily** (no load; 50 mg prophylaxis) • **anidulafungin 200 mg load → 100 mg daily** (non-enzymatic degradation) • **rezafungin 400 mg once, then 200 mg once weekly**.
- **No renal adjustment** • minimal DDIs (caspofungin + cyclosporine ↑ LFTs; micafungin ↑ sirolimus/nifedipine slightly).
- **Poor penetration** into **urine, CSF, eye** → **not for UTIs, meningitis, endophthalmitis**.
- ADRs: **histamine-mediated infusion reactions** (flushing — slow infusion), ↑ LFTs, phlebitis, hypersensitivity • very well tolerated.
- **C. parapsilosis** has higher echinocandin MICs (fluconazole preferred if susceptible).

Other antifungals

- **Flucytosine** 25 mg/kg PO q6h: **bone marrow suppression** (dose-related), hepatotoxicity, GI • **renal adjustment** • **TDM: peak 30–80 mcg/mL** • used **with AmB** (cryptococcal meningitis, Candida CNS/endocarditis).

- **Terbinafine** 250 mg daily: **hepatotoxicity** (baseline LFTs), **taste/smell disturbance**, rash/SJS, depression • **strong CYP2D6 inhibitor** (↑ metoprolol, TCAs, ↓ codeine/tramadol/tamoxifen activation) • onychomycosis: **6 weeks fingernails, 12 weeks toenails**.
- **Griseofulvin** (microsize 20–25 mg/kg/day, ultramicrosize 10–15 mg/kg/day): **take with fatty meal** (↑ absorption) • **tinea capitis 6–12 weeks** • ADRs: **photosensitivity**, headache, hepatotoxicity, GI, **disulfiram-like reaction** • **CYP inducer** → ↓ **warfarin** and **oral contraceptives** • **teratogenic** (CI pregnancy; men avoid conceiving 6 months) • CI porphyria, hepatic failure.
- **Nystatin**: not absorbed orally → topical/oral thrush (swish and swallow), cutaneous candidiasis • too toxic IV.
- **Ibrexafungerp** (VVC) and **oteseconazole** (recurrent VVC): **teratogenic** → avoid in pregnancy/females of reproductive potential (oteseconazole).

2. Superficial Fungal Infections

Dermatophytes (tinea / ringworm)

Infection	Site / features	Treatment
Tinea pedis (athlete's foot)	Interdigital scaling/maceration, moccasin type • predisposes to **cellulitis**	**Topical** azoles (clotrimazole, miconazole, ketoconazole, econazole), **terbinafine**, naftifine, butenafine, tolnaftate, ciclopirox × **1–4 weeks** (terbinafine shorter) • keep feet dry
Tinea corporis (ringworm)	Annular scaly plaque with **central clearing** and active border	Topical × 2–4 weeks (continue 1–2 weeks after clearing) • extensive → oral terbinafine/itraconazole/fluconazole
Tinea cruris (jock itch)	Groin (spares scrotum — candida involves scrotum)	Topical azoles/allylamines • treat concurrent tinea pedis
Tinea capitis	Scalp (children) — scaling, alopecia, kerion	**Oral therapy required** (topicals don't penetrate hair follicles): **griseofulvin 6–12 weeks** (Microsporum) or **terbinafine 4–6 weeks** (Trichophyton) • adjunct selenium sulfide/ketoconazole shampoo
Onychomycosis (tinea unguium)	Thick, discolored, brittle nails (toenails)	**Oral terbinafine 250 mg daily** (**6 weeks fingernails, 12 weeks toenails**) — most effective • **itraconazole** 200 mg daily × 12 weeks or pulse (200 mg BID 1 week/month) • topical **efinaconazole 10%**, tavaborole, **ciclopirox 8% nail lacquer** (mild, 48 weeks) • confirm with KOH/culture • check LFTs before oral therapy
Tinea versicolor (pityriasis versicolor — **Malassezia**, not a dermatophyte)	Hypo/hyperpigmented scaly macules on trunk	**Selenium sulfide 2.5%**, **ketoconazole 2% shampoo/cream**, topical azoles, terbinafine solution • extensive/recurrent → oral fluconazole or itraconazole (**not oral terbinafine or griseofulvin**)

Mucocutaneous candidiasis

Infection	Treatment
Oropharyngeal (thrush) — mild	**Clotrimazole troches 10 mg 5 times daily**, **miconazole buccal tablet 50 mg daily**, or **nystatin suspension** 100,000 units/mL, 4–6 mL QID (swish and swallow) × **7–14 days** • inhaled corticosteroid users → rinse mouth
Oropharyngeal — moderate–severe	**Fluconazole 100–200 mg PO daily × 7–14 days** • refractory: itraconazole solution, posaconazole, voriconazole, echinocandin, AmB
Esophageal candidiasis (HIV, odynophagia)	**Systemic therapy required**: **fluconazole 200–400 mg (3–6 mg/kg) daily × 14–21 days** • IV echinocandin if unable to swallow • refractory: itraconazole, voriconazole, posaconazole, AmB
Vulvovaginal candidiasis (VVC) — uncomplicated	**Single oral fluconazole 150 mg or topical azole** (clotrimazole, miconazole, terconazole, tioconazole) × 1–7 days (OTC) • **pregnancy → topical azole × 7 days** (avoid oral fluconazole) • partners not treated
VVC — complicated (severe, immunocompromised, uncontrolled diabetes)	**Fluconazole 150 mg every 72 h × 3 doses** or topical × 7–14 days
Recurrent VVC (≥ 3 episodes/year)	Induction (topical 7–14 days or fluconazole 150 mg q72h × 3) → **maintenance fluconazole 150 mg weekly × 6 months** • **oteseconazole** (non-reproductive potential) • **C. glabrata → boric acid 600 mg intravaginal daily × 14 days** (toxic if swallowed) or nystatin/flucytosine vaginal
Cutaneous/intertrigo, diaper candidiasis	Topical azoles or nystatin, keep area dry
Candiduria	**Asymptomatic → no treatment** (remove catheter) — except **neutropenia**, very-low-birth-weight infants, or **urologic procedure** • symptomatic cystitis → **fluconazole 200 mg daily × 2 weeks** (achieves high urine levels) • **C. glabrata/krusei** → AmB deoxycholate or flucytosine • **echinocandins and voriconazole/posaconazole/isavuconazole don't reach urine**

3. Invasive Candidiasis and Candidemia (IDSA 2016)

- **Risk factors: central venous catheters**, **broad-spectrum antibiotics**, **TPN**, ICU stay, abdominal surgery/GI perforation, **neutropenia**, chemotherapy, transplant, dialysis, burns, Candida colonization at multiple sites, prematurity.
- **Species: C. albicans** (most common, fluconazole-susceptible) • **C. glabrata** (↓ fluconazole susceptibility — elderly, prior azole) • **C. parapsilosis** (catheters, neonates; ↑ echinocandin MIC) • **C. tropicalis** (neutropenia) • **C. krusei** (**fluconazole-resistant**) • **C. auris** (multidrug-resistant, outbreaks, persists on surfaces → **contact precautions**, special lab ID).
- **Candida in blood cultures is never a contaminant** → always treat • β-D-glucan (non-specific) assists diagnosis.

Step	Recommendation
Initial therapy (non-neutropenic)	**Echinocandin** (caspofungin 70 → 50 mg, micafungin 100 mg, anidulafungin 200 → 100 mg) = **first-line** • **fluconazole 800 mg (12 mg/kg) load → 400 mg (6 mg/kg) daily** = acceptable alternative for **non-critically ill** patients unlikely to have fluconazole-resistant species • L-AmB 3–5 mg/kg if intolerance/resistance
Step-down	Switch to **fluconazole** after **5–7 days** if clinically stable, isolate **susceptible**, and repeat blood cultures negative • C. glabrata: high-dose fluconazole 800 mg or voriconazole only if susceptible • C. krusei: voriconazole
Catheter	**Remove central venous catheters** as early as possible (when source)
Eye exam	**Dilated ophthalmologic exam within the first week** (candida endophthalmitis) • neutropenic: after count recovery
Follow-up cultures	Blood cultures daily or every other day until clearance
Duration	**14 days after the first negative blood culture** and resolution of symptoms (without metastatic complications)
Neutropenic patients	**Echinocandin** or **L-AmB** • fluconazole for non-critically ill without prior azole • voriconazole if mold coverage needed

Step	Recommendation
Specific sites	**CNS**: L-AmB ± flucytosine → fluconazole • **endocarditis**: L-AmB ± flucytosine or high-dose echinocandin + **valve replacement** → long-term fluconazole • **endophthalmitis**: fluconazole/voriconazole (penetrate eye) ± intravitreal AmB/voriconazole (**not echinocandins**) • osteomyelitis 6–12 months
C. auris	**Echinocandin** first-line (often fluconazole-resistant, some AmB-resistant) • infection control
Prophylaxis	**Fluconazole**: high-risk ICU units (high incidence), liver/pancreas/small bowel transplant, HSCT • **posaconazole** (mold-active) for **AML/MDS with neutropenia** and **GVHD** • micafungin during neutropenia (HSCT)
Empiric therapy in ICU	Consider in critically ill with risk factors and unexplained fever (echinocandin) → stop at 4–5 days if no response/negative workup

4. Invasive Mold and Endemic Infections

Aspergillosis (IDSA 2016)

- **Aspergillus fumigatus** (inhaled conidia) • **invasive pulmonary aspergillosis** in **prolonged neutropenia**, HSCT/solid organ transplant, high-dose corticosteroids, GVHD, severe influenza/COVID ICU patients.
- **Diagnosis:** CT chest — nodules with **halo sign** (early), air-crescent sign (late) • **serum/BAL galactomannan** (false positives with some β-lactams historically) • β-D-glucan • culture/histology.
- **Primary therapy: voriconazole** (6 mg/kg IV q12h × 2 doses → 4 mg/kg q12h; PO 200–300 mg q12h) **or isavuconazole** (fewer ADRs) or posaconazole • **alternatives: L-AmB 3–5 mg/kg • salvage:** echinocandins, combination voriconazole + anidulafungin (severe).
- **Duration ≥ 6–12 weeks** (until lesions resolve and immunosuppression reversed) • reduce immunosuppression • **TDM for voriconazole/posaconazole**.
- **Prophylaxis: posaconazole** (AML/MDS with neutropenia, GVHD), voriconazole, isavuconazole.
- **Allergic bronchopulmonary aspergillosis (ABPA)** (asthma, cystic fibrosis): **corticosteroids + itraconazole** • **aspergilloma** (fungus ball in cavity) → surgery if hemoptysis • **chronic pulmonary aspergillosis** → oral azole ≥ 6 months.

Mucormycosis (zygomycosis)

- **Rhizopus, Mucor** • **rhino-orbital-cerebral** disease (black eschar on palate/nasal turbinates) in **diabetic ketoacidosis**, neutropenia, steroids, **deferoxamine** therapy, iron overload, transplant (and COVID-associated).
- **Treatment: urgent surgical debridement** + **L-AmB 5–10 mg/kg/day** (first-line) + correct underlying condition (DKA, stop steroids/deferoxamine) • **step-down/salvage: isavuconazole or posaconazole** • **voriconazole and echinocandins have NO activity** (breakthrough on voriconazole prophylaxis).

Endemic mycoses (dimorphic fungi, inhaled)

Fungus	Epidemiology / features	Treatment
Histoplasma capsulatum	Ohio/Mississippi river valleys • **bird/bat droppings** (caves) • pulmonary, disseminated in HIV (CD4 < 150)	Mild–moderate: **itraconazole 200 mg TID × 3 days → 200 mg daily–BID × 6–12 weeks** • severe/disseminated: **L-AmB 1–2 weeks** → itraconazole ≥ 12 months
Blastomyces dermatitidis	Great Lakes, Ohio/Mississippi • pulmonary, skin (verrucous), bone, GU	Mild–moderate: **itraconazole 6–12 months** • severe: L-AmB → itraconazole
Coccidioides immitis	Southwest US (California — San Joaquin "valley fever"), Mexico • dust	**Fluconazole 400–800 mg** (or itraconazole) • severe: L-AmB • **meningitis → lifelong fluconazole**
Sporothrix schenckii	Rose gardeners (thorn), lymphocutaneous nodules	**Itraconazole** 3–6 months • severe: L-AmB
Paracoccidioides, Talaromyces	Latin America / Southeast Asia (HIV)	Itraconazole / L-AmB → itraconazole

Cryptococcosis

- **Cryptococcus neoformans** (pigeon droppings) → **meningitis** in **HIV (CD4 < 100)** or transplant • subacute headache, fever • **↑ opening pressure** • **CSF/serum cryptococcal antigen (CrAg)**, India ink.
- **Induction (2 weeks): liposomal AmB 3–4 mg/kg/day + flucytosine 25 mg/kg q6h** (WHO preferred: **single dose L-AmB 10 mg/kg** + 14 days flucytosine + fluconazole 1,200 mg) • **consolidation: fluconazole 400–800 mg daily × 8 weeks • maintenance: fluconazole 200 mg daily ≥ 1 year** (stop when CD4 ≥ 100 and viral load suppressed on ART).
- **Manage ↑ intracranial pressure with serial therapeutic lumbar punctures** (not acetazolamide, mannitol, or steroids).
- **Delay ART for 4–6 weeks** after starting antifungals (IRIS → ↑ mortality) • non-meningeal mild pulmonary → fluconazole 400 mg × 6–12 months.
- **Asymptomatic serum CrAg+ in HIV** → preemptive fluconazole.

Pneumocystis jirovecii pneumonia (PCP/PJP)

- Fungus • **HIV (CD4 < 200)**, transplant, high-dose steroids, chemotherapy • subacute dyspnea, dry cough, **hypoxemia**, bilateral interstitial infiltrates, ↑ LDH, β-D-glucan ↑ • diagnosis: induced sputum/BAL (DFA, PCR).
- **Treatment: TMP-SMX 15–20 mg/kg/day (TMP component) IV/PO divided q6–8h × 21 days** (drug of choice).
- **Adjunctive prednisone** if **PaO2 < 70 mmHg** or **A-a gradient ≥ 35** — start within 72 h: **40 mg BID days 1–5 → 40 mg daily days 6–10 → 20 mg daily days 11–21**.
- **Alternatives (sulfa allergy): primaquine + clindamycin** (check G6PD), **IV pentamidine** 4 mg/kg/day (severe; ADRs: hypoglycemia then hyperglycemia, pancreatitis, nephrotoxicity, hypotension, QT), **atovaquone** 750 mg BID with food (mild–moderate), **dapsone + trimethoprim** (mild–moderate; G6PD).
- **Prophylaxis** (HIV CD4 < 200 or oropharyngeal candidiasis; transplant; prednisone ≥ 20 mg ≥ 4 weeks): **TMP-SMX 1 DS daily** (or 1 SS daily/1 DS 3×/week) • alternatives: **dapsone 100 mg daily** (G6PD), **atovaquone 1,500 mg daily**, **aerosolized pentamidine 300 mg monthly** • stop when CD4 > 200 for ≥ 3 months on ART.

5. Antifungal Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
AmB deoxycholate	**0.5–1 mg/kg/day** (max 1.5)	**Infusion reactions, nephrotoxicity, ↓ K, ↓ Mg**, RTA, anemia	Premedicate • NS loading • **D5W only** • fatal if overdosed
Liposomal AmB	**3–5 mg/kg/day** (mucor 5–10; crypto 3–4)	Less nephrotoxic, infusion reactions	Mucor, crypto, molds, CNS

Drug	Dose	Key ADRs	Key notes / interactions
Fluconazole	VVC 150 mg once • thrush 100–200 mg • candidemia **800 → 400 mg** • esophageal 200–400 mg	Hepatotoxicity, QT, GI	**Renal adjust** (CrCl ≤ 50) • IV = PO • ↑ **warfarin, phenytoin** (2C9) • no C. krusei, molds
Itraconazole	200 mg daily–BID (load TID × 3 d)	**HF (boxed)**, hepatotoxicity	Capsule with food + acid (no PPIs) • solution empty stomach • strong 3A4 inhibitor
Voriconazole	6 mg/kg IV q12h × 2 → 4 mg/kg q12h • PO 200–300 mg q12h	**Visual changes**, hallucinations, **photosensitivity/SCC**, hepatotoxicity, QT, periostitis	**Aspergillus DOC** • TDM 1–5.5 • CYP2C19 • IV vehicle CrCl < 50 • CI sirolimus, rifampin, carbamazepine • no Mucor
Posaconazole	DR tab 300 mg BID day 1 → 300 mg daily (with food)	Hepatotoxicity, QT, ↓ K, HTN	Prophylaxis (AML, GVHD) • Mucor step-down • strong 3A4 inhibitor
Isavuconazole	372 mg q8h × 6 doses → 372 mg daily (IV/PO)	**QT shortening**, hepatotoxicity (less)	Aspergillus, Mucor • moderate 3A4 inhibitor
Caspofungin	**70 mg → 50 mg daily** IV	Infusion reaction, ↑ LFTs	Hepatic adjust • ↑ with inducers • not for UTI/CNS/eye
Micafungin	**100 mg daily** (EC 150; prophylaxis 50)	↑ LFTs, infusion reaction	No load • no renal adjust
Anidulafungin	**200 mg → 100 mg daily**	↑ LFTs	No hepatic/renal adjust
Rezafungin	400 mg once → 200 mg weekly	Infusion reactions, photosensitivity	Once weekly
Flucytosine	25 mg/kg PO q6h	**Bone marrow suppression**, hepatotoxicity	With AmB (crypto) • TDM peak 30–80 • renal adjust
Terbinafine	**250 mg daily** (6 wk fingernail, 12 wk toenail)	**Hepatotoxicity**, taste/smell loss, SJS	**CYP2D6 inhibitor** • LFTs baseline
Griseofulvin	Microsize 20–25 mg/kg/day (adult 500–1,000 mg) with fatty meal	Photosensitivity, headache, hepatotoxicity	**Inducer** (↓ OCs, warfarin) • teratogenic • tinea capitis
Nystatin	Suspension 400,000–600,000 units QID (swish and swallow)	GI (minimal)	Not absorbed • thrush
Clotrimazole troche	10 mg 5×/day × 7–14 d	Nausea	Thrush (mild)
TMP-SMX (PCP)	**15–20 mg/kg/day TMP** × 21 d • prophylaxis 1 DS daily	Rash/SJS, hyperkalemia, marrow suppression, ↑ SCr	+ prednisone if PaO2 < 70 • ↑ INR
Pentamidine IV	4 mg/kg daily	**Hypo/hyperglycemia**, pancreatitis, nephrotoxicity, hypotension, QT	PCP alternative (severe)
Atovaquone	Treatment 750 mg BID • prophylaxis 1,500 mg daily (with fatty food)	GI, rash	Mild PCP, prophylaxis
Dapsone	100 mg daily	**Hemolysis (G6PD)**, **methemoglobinemia**	PCP prophylaxis • check G6PD

PART B — IMMUNIZATIONS

6. Vaccine Principles

- **Active immunity** = vaccines (body makes antibodies; long-lasting) • **passive immunity** = immune globulins/monoclonal antibodies (HBIG, RIG, TIG, VariZIG, IVIG, nirsevimab — immediate, temporary).
- **Vaccine types: live attenuated** • **inactivated** (whole/split) • **subunit/recombinant** (HBV, HPV, recombinant zoster, RSV) • **toxoid** (tetanus, diphtheria) • **polysaccharide** (PPSV23 — T-cell independent, poor in < 2 years, no booster effect) • **conjugate** (PCV, Hib, MenACWY — T-cell dependent, works in infants) • **mRNA** (COVID-19, RSV mResvia) • viral vector.
- **VAERS** (reporting adverse events) • **Vaccine Injury Compensation Program** • give the **Vaccine Information Statement (VIS)** before each dose • document lot number, site, date.

Live attenuated vaccines (high-yield list)

- **MMR**, **varicella** (Varivax), **MMRV** (ProQuad), **rotavirus** (oral), **LAIV** (FluMist intranasal, ages 2–49), **yellow fever**, **oral typhoid** (Ty21a), **BCG**, oral cholera (Vaxchora), smallpox (ACAM2000), oral polio (not used in US) • (live zoster vaccine Zostavax **discontinued** — only recombinant Shingrix now).
- **Contraindicated in pregnancy** (theoretical fetal risk → avoid pregnancy for **4 weeks** after MMR/varicella) and **severe immunosuppression**: chemotherapy, HIV with **CD4 < 200** (or < 15% in children), **prednisone ≥ 20 mg/day (or ≥ 2 mg/kg) for ≥ 14 days**, biologics (TNF inhibitors), transplant, congenital immunodeficiency • household contacts of immunocompromised persons **can** receive MMR, varicella, rotavirus (hand hygiene after diapers).
- **Spacing:** two **live parenteral/intranasal** vaccines → **same day or ≥ 28 days apart** • inactivated vaccines → any interval with any vaccine.
- **Antibody-containing products** (IVIG, blood products) **delay MMR and varicella** by 3–11 months (antibodies neutralize the vaccine virus) — not for inactivated vaccines.
- **TB skin test with MMR/varicella:** give **on the same day** or **wait ≥ 4 weeks** (live vaccine suppresses TST → false negative).
- Give live vaccines **≥ 4 weeks before** starting immunosuppression • inactivated **≥ 2 weeks before**.

Contraindications and precautions (all vaccines)

- **Contraindication: severe allergic reaction (anaphylaxis)** to a previous dose or vaccine component.
- **Precaution: moderate or severe acute illness** with or without fever → defer • history of **GBS within 6 weeks** of a prior influenza or tetanus vaccine.
- **NOT contraindications:** mild illness (cold, low fever), current antibiotics (except oral typhoid/live bacterial), breastfeeding, prematurity, recent exposure to infection, family history of adverse events, **egg allergy for flu vaccine** (any flu vaccine at any severity, no extra observation since 2023), mild local reactions.
- **Specific: encephalopathy within 7 days** of DTaP/Tdap → contraindication to further pertussis vaccine • **intussusception history or SCID** → no rotavirus • **severe egg allergy** → yellow fever (not flu) • **yeast allergy** → hepatitis B, HPV • **gelatin** (MMR, varicella, yellow fever) • **neomycin** (MMR, varicella, IPV) • **latex** (some prefilled syringe caps).

Administration and storage

- **IM** (most inactivated vaccines): **deltoid** in adults/children ≥ 3 years, **anterolateral thigh (vastus lateralis)** in infants • needle 1–1.5 inch (adults) • **never gluteal** (↓ immunogenicity — esp. hepatitis B and rabies).
- **Subcutaneous: MMR, varicella, MMRV**, yellow fever (IPV and PPSV23 SC or IM) • **intranasal:** LAIV • **oral:** rotavirus, oral typhoid, cholera • **intradermal:** BCG, TST.
- **Storage:** most vaccines **refrigerated 2–8 °C** (do not freeze) • **varicella, MMRV frozen** (−50 to −15 °C) • MMR fridge or freezer • monitor temperatures; never store in fridge door.
- **Syncope** (adolescents) → observe **15 minutes** seated • anaphylaxis kit (epinephrine).
- Simultaneous administration of all age-appropriate vaccines is safe • **don't restart a series** if interrupted — continue where stopped ("a dose counts").

7. Childhood and Adolescent Schedule (birth–18 years)

Vaccine	Schedule	Notes
Hepatitis B	**Birth** (within 24 h), 1–2 months, 6–18 months	HBsAg+ mother → HepB vaccine + **HBIG within 12 h** of birth
Rotavirus (RV1 oral 2 doses / RV5 3 doses)	2, 4, (6) months	**First dose by 14 weeks 6 days**, **max age 8 months 0 days** • risk: intussusception • CI SCID
DTaP	**2, 4, 6, 15–18 months, 4–6 years** (5 doses)	Fever, local reactions; encephalopathy → no further pertussis vaccine
Hib (conjugate)	2, 4, (6), 12–15 months	PRP-OMP needs no 6-month dose
Pneumococcal conjugate (PCV15 or PCV20)	2, 4, 6, 12–15 months	PPSV23 later for high-risk conditions if PCV15 used
IPV (inactivated polio)	2, 4, 6–18 months, 4–6 years	SC or IM
Influenza	**Every year ≥ 6 months**	**2 doses (4 weeks apart)** for children 6 months–8 years receiving it for the first time
MMR	**12–15 months, 4–6 years**	Infants 6–11 months before international travel → 1 dose (doesn't count) • febrile seizures (MMRV > separate MMR+V for first dose)
Varicella	**12–15 months, 4–6 years**	Live • SC
Hepatitis A	2 doses at 12–23 months (6 months apart)	Inactivated
Tdap	**11–12 years**	Then Td/Tdap every 10 years
HPV (9-valent)	**11–12 years** (can start at 9)	**2 doses (0, 6–12 months) if started before 15th birthday • 3 doses (0, 1–2, 6 months)** if started ≥ 15 years or immunocompromised • through age 26
MenACWY (conjugate)	**11–12 years + booster at 16**	High-risk from 2 months
MenB	16–23 years (shared decision; preferred 16–18)	High-risk ≥ 10 years (asplenia, complement deficiency)
RSV monoclonal antibody (**nirsevimab**, clesrovimab)	Infants < 8 months entering first RSV season (if mother not vaccinated ≥ 14 days before birth)	Single IM dose • high-risk 8–19 months second season
COVID-19	≥ 6 months per current recommendations	Recommendations changed in 2025 (individual/shared decision-making) — follow course reference
BCG (Saudi schedule/high-prevalence countries)	At birth	Live • not in US routine schedule

8. Adult Immunization Schedule (≥ 19 years)

Vaccine	Who / schedule
Influenza (inactivated or recombinant)	**All adults annually • ≥ 65: preferentially high-dose (Fluzone HD), adjuvanted (Fluad), or recombinant (Flublok) • LAIV** only healthy non-pregnant 2–49 years (not immunocompromised, not caregivers of severely immunocompromised, not with recent antivirals)
Tdap / Td	**One dose of Tdap** if not received as adolescent → **Td or Tdap every 10 years • Tdap during EVERY pregnancy at 27–36 weeks** (early in that window) • wound management (below)

Vaccine	Who / schedule
MMR	Adults born **≥ 1957** without evidence of immunity → **1 dose** • **2 doses** (4 weeks apart) for **healthcare workers**, college students, international travelers • **not in pregnancy** (vaccinate postpartum) • HIV CD4 ≥ 200 OK
Varicella	Adults without evidence of immunity (born before 1980 in US considered immune — except HCWs, pregnant, immunocompromised) → **2 doses 4–8 weeks apart** • not in pregnancy
Zoster — recombinant (RZV, Shingrix)	**All adults ≥ 50 years**: **2 doses 2–6 months apart** IM • **≥ 19 years immunocompromised** (2 doses 1–2 months apart) • give regardless of prior shingles, prior Zostavax, or varicella history • non-live (OK in immunocompromised) • ADRs: **reactogenic** (sore arm, myalgia, fatigue, fever 2–3 days)
HPV	Through **age 26** (catch-up) • **27–45 shared clinical decision-making** • not in pregnancy (delay)
Pneumococcal (ACIP 2024)	**All adults ≥ 50 years** and **19–49 with risk conditions** (chronic heart/lung/liver/renal disease, diabetes, alcoholism, smoking, immunocompromise, asplenia, CSF leak, cochlear implant, HIV) → **PCV20 or PCV21 alone** (single dose) **OR PCV15 followed by PPSV23 ≥ 1 year later** (≥ 8 weeks if immunocompromised, CSF leak, cochlear implant) • previously PPSV23 only → PCV20/21 ≥ 1 year later
Hepatitis B	**All adults 19–59 years** • **≥ 60 with risk factors** (diabetes, HCW, dialysis, HIV, HCV, IDU, multiple partners, MSM, chronic liver disease, incarceration, travel) or anyone requesting • **Engerix-B/Recombivax HB: 3 doses (0, 1, 6 months)** • **Heplisav-B: 2 doses (0, 1 month)** • **dialysis: high-dose (40 mcg)** • check **anti-HBs ≥ 10 mIU/mL** 1–2 months after series in HCWs, dialysis, immunocompromised, infants of HBsAg+ mothers
Hepatitis A	Risk: travel to endemic areas, **chronic liver disease**, MSM, IDU, homelessness, HIV, clotting factor disorders, close contact with adoptee, pregnancy at risk • **2 doses (0, 6–12 months)** • Twinrix (HepA-HepB) 3 doses 0, 1, 6 months
Meningococcal ACWY	**Asplenia**, **complement deficiency/eculizumab or ravulizumab**, HIV (2-dose primary), microbiologists, travel to **meningitis belt**, **Hajj/Umrah** (required), military recruits, first-year college students in dorms • boosters every 5 years if risk persists
Meningococcal B	Asplenia, complement deficiency/complement inhibitors, microbiologists, outbreaks • (MenABCWY pentavalent options)
RSV (Arexvy, Abrysvo, mResvia)	**Single dose** for **all adults ≥ 75** and **50–74 at increased risk** (chronic heart/lung/kidney disease, diabetes with complications, immunocompromise, obesity BMI ≥ 40, nursing home) • ideally late summer/early fall • **pregnancy: Abrysvo at 32–36 weeks** (September–January) to protect the infant
COVID-19	Current-season vaccine — 2025 ACIP: individual-based (shared) decision-making; FDA approvals focus on ≥ 65 and high-risk • follow current/course guidance
Hib	Asplenia (1 dose if not previously vaccinated), HSCT (3 doses)

9. Special Populations

Population	Recommended / avoid
Pregnancy	**Recommended: inactivated influenza** (any trimester), **Tdap (27–36 weeks, every pregnancy)**, **RSV (Abrysvo 32–36 weeks, seasonal)**, COVID-19, hepatitis B/A if at risk • **Avoid: MMR, varicella, LAIV** (live), **HPV** (delay), recombinant zoster (delay) • breastfeeding: all vaccines OK (except smallpox, yellow fever caution)
Immunocompromised (chemotherapy, transplant, high-dose steroids, biologics, HIV CD4 < 200)	**No live vaccines** • inactivated vaccines are safe (may have ↓ response; some need extra doses) • recombinant zoster from age 19 • PCV20/21 • give vaccines before immunosuppression (live ≥ 4 weeks, inactivated ≥ 2 weeks)
HIV	CD4 ≥ 200: MMR, varicella OK • all inactivated vaccines • MenACWY 2-dose primary • PCV20/21 • Hep B, Hep A • Shingrix • HPV through 26
Asplenia / sickle cell	**Pneumococcal**, **MenACWY**, **MenB**, **Hib** (encapsulated organisms) — **≥ 14 days before elective splenectomy** (or ≥ 14 days after emergency splenectomy) • annual flu

Population	Recommended / avoid
Healthcare workers	**Hepatitis B** (check anti-HBs), **MMR (2 doses)**, **varicella (2 doses)**, **influenza annually**, **Tdap**, MenACWY/MenB (microbiologists) • baseline TB testing
Diabetes	Hepatitis B (19–59; ≥ 60 consider), pneumococcal, influenza, Tdap, zoster ≥ 50, RSV (if complications), COVID
Chronic kidney disease / dialysis	**High-dose hepatitis B** (40 mcg) with anti-HBs monitoring yearly, pneumococcal, influenza
Chronic liver disease	**Hepatitis A and B**, pneumococcal, influenza
Older adults ≥ 65	**High-dose/adjuvanted/recombinant influenza**, **pneumococcal**, **Shingrix**, **RSV (≥ 75 all; 50–74 at risk)**, Td/Tdap, COVID
Travelers	**Hepatitis A**, **typhoid** (oral Ty21a live — 1 capsule every other day × 4 doses, 1 h before a meal with cool liquid, avoid antibiotics; or injectable Vi polysaccharide IM ≥ 2 years, booster every 2 years), **yellow fever** (live, ≥ 9 months, certificate valid for life; CI egg anaphylaxis, thymus disorder, immunosuppression; caution ≥ 60), **meningococcal** (Hajj/Umrah, meningitis belt), Japanese encephalitis, rabies pre-exposure (2 doses days 0 and 7), polio booster, cholera
Hajj and Umrah (Saudi Arabia requirements)	**Quadrivalent meningococcal (ACWY) vaccine** required ≥ 10 days before arrival (conjugate valid 5 years, polysaccharide 3 years) • seasonal influenza, COVID-19 and polio (for travelers from polio-affected countries) recommended/required per current rules • yellow fever certificate if arriving from endemic countries

10. Post-Exposure Prophylaxis (PEP)

Tetanus prophylaxis in wound management

Tetanus vaccine history	Clean, minor wound	All other wounds (dirty, punctures, burns, crush)
Unknown or < 3 doses	**Tdap/Td** (DTaP if < 7 years) — no TIG	**Tdap/Td + tetanus immune globulin (TIG) 250 units IM** (different site)
≥ 3 doses	Tdap/Td only if **≥ 10 years** since last dose	Tdap/Td only if **≥ 5 years** since last dose (no TIG)

Rabies PEP

- **Wound care** first: wash thoroughly with soap and water for ~15 min + povidone-iodine.
- **Not previously vaccinated: human rabies immune globulin (HRIG) 20 IU/kg** on day 0 — **infiltrate as much as possible around the wound**, rest IM at a site distant from vaccine (**never in the same syringe or site** as vaccine; don't exceed dose — suppresses response) + **rabies vaccine 1 mL IM in deltoid on days 0, 3, 7, and 14** (+ **day 28** if immunocompromised) • **never gluteal**.
- **Previously vaccinated:** vaccine on **days 0 and 3 only — no HRIG**.
- Indications: bites/scratches from bats (any contact), wild carnivores (raccoons, skunks, foxes), unvaccinated/unobservable dogs/cats • healthy dog/cat can be observed 10 days.

Other PEP

- **Hepatitis B exposure** (needlestick/sexual): unvaccinated → **HBIG + start HepB vaccine** (within 24 h, ≤ 7 days) • vaccinated responder (anti-HBs ≥ 10) → no treatment • non-responder → HBIG ×2 or HBIG + revaccination.
- **Varicella exposure:** susceptible healthy → **varicella vaccine within 3–5 days** • high-risk (immunocompromised, pregnant, neonates) → **VariZIG within 10 days** (or IVIG) ± acyclovir.
- **Measles exposure: MMR within 72 h** or **immune globulin within 6 days** (infants < 6 months, pregnant, immunocompromised).
- **Hepatitis A exposure: HepA vaccine and/or immune globulin within 2 weeks** (IG for < 12 months, immunocompromised, chronic liver disease, ≥ 40 years per risk).
- **Pertussis:** azithromycin for close contacts • **meningococcus:** rifampin/cipro/ceftriaxone • **influenza:** oseltamivir 75 mg daily × 7 days • **HIV:** 3-drug ART (e.g., bictegravir/emtricitabine/TAF) within 72 h × 28 days.

11. Vaccine Adverse Effects and Specific Notes

Vaccine	Notable adverse effects / points
All injectable vaccines	Local pain, redness, swelling, low-grade fever, **syncope** (adolescents), anaphylaxis (rare, within minutes–hours)
MMR	Fever and **rash 7–12 days** after, transient arthralgia (women), **febrile seizures** (↑ with MMRV in 12–47 months), **thrombocytopenia (ITP)** • **no link to autism**
Varicella	Injection-site rash, mild varicella-like rash (can transmit rarely — avoid contact with immunocompromised if rash)
Rotavirus	**Intussusception** (small risk, first week after dose 1) • irritability, diarrhea
DTaP / Tdap	Fever, fussiness, swelling (extensive limb swelling with 4th/5th dose), **prolonged crying**, rarely hypotonic-hyporesponsive episodes • encephalopathy within 7 days → contraindication to pertussis component
Influenza (inactivated)	Sore arm, low-grade fever • **GBS** (~1 per million) • **LAIV**: runny nose, wheezing (avoid in children 2–4 with asthma/wheezing)
Recombinant zoster	**Highly reactogenic** (myalgia, fatigue, fever, injection-site pain for 2–3 days) — counsel to ensure 2nd dose • small GBS signal
RSV vaccines	Injection-site pain, fatigue • small risk of **GBS** (older adults) • preterm birth signal with Abrysvo → give only at 32–36 weeks
mRNA COVID-19	Fatigue, headache, fever • **myocarditis/pericarditis** (young males, after 2nd dose)
HPV	Syncope, local reactions • prevents cervical, anal, oropharyngeal cancers and genital warts
Hepatitis B	Local reactions • Heplisav-B (CpG adjuvant) 2 doses
Yellow fever	Vaccine-associated viscerotropic/neurotropic disease (↑ in ≥ 60, thymus disease)
BCG	Local ulcer/scar, lymphadenitis, disseminated BCG in immunocompromised • false-positive TST

12. Immunization Product Summary

Product	Dose / schedule	Key ADRs	Key notes
Influenza (IIV/RIV)	0.5 mL IM annually (≥ 65 high-dose/adjuvanted/recombinant)	Sore arm, GBS (rare)	Egg allergy OK • 2 doses first time 6 mo–8 y
LAIV (FluMist)	0.2 mL intranasal (2–49 y)	Rhinorrhea, wheeze	**Live** • not in pregnancy/immunocompromised • antivirals ↓ efficacy
Tdap / Td	0.5 mL IM • Tdap once, then every 10 years • each pregnancy 27–36 wk	Local, fever	Wound: 5 years (dirty) / 10 years (clean)
MMR	0.5 mL SC • 12–15 mo and 4–6 y • adults 1–2 doses	Fever/rash, febrile seizures, ITP	**Live** • avoid pregnancy 4 wk • TST same day or 4 wk later
Varicella	0.5 mL SC • 2 doses	Rash	**Live** • frozen storage
Recombinant zoster (Shingrix)	0.5 mL IM • 2 doses 2–6 months apart (≥ 50; ≥ 19 immunocompromised)	**Reactogenic**	Non-live • after shingles/Zostavax
PCV20 / PCV21	Single dose (≥ 50 or 19–49 at risk)	Local	No PPSV23 needed after
PCV15 + PPSV23	PCV15 → PPSV23 ≥ 1 year (≥ 8 weeks if immunocompromised)	Local	PPSV23 polysaccharide

Product	Dose / schedule	Key ADRs	Key notes
Hepatitis B	Engerix/Recombivax 0, 1, 6 months • Heplisav-B 0, 1 month • dialysis 40 mcg	Local	All 19–59 • anti-HBs ≥ 10 check
Hepatitis A	2 doses 0, 6–12 months	Local	Liver disease, travel, MSM, IDU
HPV-9	2 doses (start < 15) or 3 doses (≥ 15, immunocompromised)	Syncope	Through 26 (27–45 shared)
MenACWY	11–12 y + 16 y booster • high risk/Hajj	Local	Required for Hajj/Umrah
MenB	2–3 doses (product-specific)	Local, fever	Asplenia, complement inhibitors
RSV vaccine	Single dose ≥ 75 / 50–74 at risk • pregnancy Abrysvo 32–36 wk	GBS (rare)	Not repeated yearly
Nirsevimab	50 mg (< 5 kg) or 100 mg (≥ 5 kg) IM once	Rash, injection-site	Infant RSV protection (passive)
Rotavirus	Oral 2–3 doses (start ≤ 14 wk 6 d, finish by 8 mo)	**Intussusception**	**Live** • CI SCID
HRIG + rabies vaccine	HRIG 20 IU/kg day 0 (wound) • vaccine days 0, 3, 7, 14 (± 28)	Local, headache	Deltoid, never gluteal • previously vaccinated: days 0, 3 only
TIG	250 units IM	Local	Dirty wound + < 3 tetanus doses/unknown
HBIG	0.06 mL/kg IM (newborn 0.5 mL)	Local	HBV PEP, newborns of HBsAg+ mothers (within 12 h)
VariZIG	125 units/10 kg (max 625) IM within 10 days	Local	High-risk varicella exposure
Yellow fever	0.5 mL SC once (lifelong)	Viscerotropic disease (rare)	**Live** • ≥ 9 months • egg anaphylaxis CI
Typhoid	Oral Ty21a every other day × 4 caps • Vi IM every 2 years	GI / local	Oral is live — avoid antibiotics

Exam Traps and Frequently Tested Points

1. **AmB** binds ergosterol • **azoles** inhibit 14-α-demethylase • **echinocandins** inhibit β-1,3-D-glucan synthase (cell wall) • **terbinafine** squalene epoxidase • **griseofulvin** microtubules • **flucytosine** → 5-FU.
2. **AmB:** infusion reactions (premedicate; meperidine for rigors), **nephrotoxicity** (saline loading), **↓ K, ↓ Mg** • conventional max **1.5 mg/kg** • D5W only • lipid forms less toxic.
3. **Fluconazole:** IV = PO, renal dose adjust, great urine/CSF penetration • **no C. krusei, no molds** • ↑ warfarin/phenytoin (2C9).
4. **Itraconazole:** capsule with food + acidic drink (no PPIs), solution empty stomach • **boxed HF** • DOC histoplasmosis/blastomycosis.
5. **Voriconazole: DOC aspergillosis** • visual disturbances, hallucinations, photosensitivity/skin cancer, periostitis • TDM (2C19) • IV vehicle avoid CrCl < 50 • **no Mucor activity** • CI sirolimus.
6. **Mucormycosis** (DKA, deferoxamine): **surgery + L-AmB** → isavuconazole/posaconazole • voriconazole and echinocandins inactive.
7. **Posaconazole**: prophylaxis in AML/MDS and GVHD • DR tablets with food • **isavuconazole shortens QT**.
8. **Echinocandins:** IV only, no renal adjust, few DDIs • **not for UTI, CNS, eye** • caspofungin 70 → 50 mg • C. parapsilosis ↑ MIC.
9. **Candidemia: echinocandin first** → step down to fluconazole after 5–7 days if stable/susceptible • **remove CVC • eye exam • 14 days after first negative culture**.

10. **C. krusei** fluconazole-resistant • **C. glabrata** reduced susceptibility • **C. auris** MDR → echinocandin + contact precautions.
11. **Candiduria:** asymptomatic → don't treat (except neutropenia, VLBW infants, urologic procedure) • symptomatic → **fluconazole 200 mg × 2 weeks**.
12. **Thrush:** clotrimazole troches, miconazole buccal, nystatin → fluconazole if moderate–severe • **esophageal → systemic fluconazole 14–21 days**.
13. **VVC: fluconazole 150 mg once** or topical azole • complicated → 150 mg q72h × 3 • recurrent → weekly × 6 months • **pregnancy → topical 7 days** • C. glabrata → boric acid.
14. **Tinea capitis and onychomycosis need oral therapy** • terbinafine 6 weeks (fingernails) / 12 weeks (toenails) • griseofulvin with fatty meal (inducer, teratogenic).
15. **Tinea versicolor** = Malassezia → selenium sulfide, ketoconazole • not oral terbinafine/griseofulvin.
16. **Cryptococcal meningitis:** L-AmB + flucytosine × 2 weeks → fluconazole 400–800 × 8 weeks → 200 mg ≥ 1 year • **serial LPs** for ICP • **delay ART 4–6 weeks**.
17. **PCP:** TMP-SMX 15–20 mg/kg/day × 21 days + **prednisone if PaO2 < 70 or A–a ≥ 35** • prophylaxis CD4 < 200: TMP-SMX DS daily (dapsone, atovaquone, pentamidine) • check G6PD for dapsone/primaquine.
18. **Live vaccines:** MMR, varicella, MMRV, rotavirus, LAIV, yellow fever, oral typhoid, BCG • **CI in pregnancy and severe immunosuppression** (CD4 < 200, pred ≥ 20 mg ≥ 14 days).
19. **Two live vaccines:** same day or ≥ 28 days apart • **TST:** same day as MMR or wait 4 weeks • IVIG/blood delays MMR/varicella.
20. **Only true contraindication for all vaccines:** anaphylaxis to prior dose/component • mild illness and antibiotics are not contraindications • **egg allergy → any flu vaccine**.
21. **Infant IM site:** vastus lateralis • adults: deltoid • **never gluteal** (HepB, rabies) • MMR/varicella SC • varicella frozen • observe 15 min (syncope).
22. **Rotavirus:** first dose by 14 weeks 6 days, last by 8 months • intussusception • **HPV:** 2 doses if started < 15 years, 3 doses if ≥ 15 or immunocompromised.
23. **HepB at birth** (+ HBIG within 12 h if mother HBsAg+) • **DTaP 5 doses** • **MMR/varicella 12–15 months + 4–6 years** • **Tdap 11–12** • **MenACWY 11–12 + 16**.
24. **Influenza:** everyone ≥ 6 months yearly • 2 doses for first-time 6 months–8 years • **≥ 65 → high-dose/adjuvanted/recombinant** • LAIV only 2–49 healthy non-pregnant.
25. **Tdap every pregnancy (27–36 weeks)** • Td/Tdap every 10 years • dirty wound: booster if ≥ 5 years; TIG if < 3 doses/unknown.
26. **Shingrix:** all ≥ 50 (≥ 19 immunocompromised) • 2 doses 2–6 months • non-live • give even after shingles/Zostavax • reactogenic.
27. **Pneumococcal (2024):** all ≥ 50 and 19–49 with risk → **PCV20 or PCV21 alone**, or **PCV15 → PPSV23 ≥ 1 year** (8 weeks if immunocompromised).
28. **Hepatitis B:** all adults 19–59 (≥ 60 with risk) • Heplisav-B 2 doses • dialysis high dose • anti-HBs ≥ 10 in HCWs.
29. **RSV:** single dose ≥ 75 and 50–74 at risk • **pregnancy Abrysvo 32–36 weeks** or infant nirsevimab.
30. **Pregnancy:** give inactivated flu, Tdap, RSV, COVID • **avoid MMR, varicella, LAIV, HPV**.
31. **Asplenia:** pneumococcal, MenACWY, MenB, Hib ≥ 14 days before splenectomy • **eculizumab/ravulizumab** → meningococcal vaccines (ACWY + B).
32. **Hajj/Umrah: MenACWY required** ≥ 10 days before arrival (conjugate valid 5 years) • yellow fever certificate from endemic countries.
33. **Rabies PEP:** wound wash → **HRIG 20 IU/kg into wound** + vaccine days **0, 3, 7, 14** (+ 28 immunocompromised) deltoid • previously vaccinated → days 0, 3 only, no HRIG.
34. **Other PEP:** HBV → HBIG + vaccine • varicella → vaccine ≤ 3–5 days or VariZIG ≤ 10 days • measles → MMR ≤ 72 h or IG ≤ 6 days • hep A → vaccine/IG ≤ 2 weeks.
35. **MMR → febrile seizures (MMRV higher), ITP, rash 7–12 days** • rotavirus → intussusception • mRNA → myocarditis (young males) • DTaP encephalopathy → no more pertussis vaccine.

Note: antifungal recommendations follow IDSA mycoses guidelines; immunization schedules follow CDC/ACIP (with 2024–2025 updates — some 2025 ACIP changes, e.g., COVID-19, differ from professional societies). If your course slides or reference use slightly different recommendations, **use your course material for the exam**.